K223: Nuclear γ-γ angular correlations

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We measured the angular γ-γ correlation of the ^{60}Co 4–2–0 cascade with two scintillation detectors in a fast–slow coincidence setup. After calibrating the amplifier gains, the SCA and CFD thresholds, and the fast- and slow-branch delays, we determined a prompt-curve resolving time of 32.35(95) ns. We measured the random-coincidence contribution at the same set of angles as the main scan and subtracted it angle by angle. We then corrected the coincidence data for geometric misalignment using the ratio of the fixed-detector and movable-detector singles rates. A fit of the corrected angular-correlation data with a free Legendre expansion yields $A_{22}^{\mathrm{exp}} = 0.090(4)$ and $A_{44}^{\mathrm{exp}} = 0.012(5)$, corresponding to the finite-solid-angle-corrected values $A_{22}^{\mathrm{true}} = 0.098(4)$ and $A_{44}^{\mathrm{true}} = 0.024(9)$. The goodness-of-fit of the fitted corrected data yields a p-value of $p = 14.7\%$ which doesn't conclusively point to statistical incompatibility with the expected 4–2–0 cascade shape, but may point to some residual effects present in the experiment. In addition, both an isotropic distribution and the alternative 2–1–0 hypothesis are strongly excluded.

I. INTRODUCTION

Angular γ-γ correlations provide direct information on nuclear spin states and transition multipolarities. If an excited nucleus emits two successive γ rays, the direction of the second photon is, in general, not independent of the direction of the first one. Instead, the coincidence rate depends on the opening angle between the two detectors, because the detection of the first transition selects a subset of intermediate nuclear states and thereby imposes a characteristic angular distribution on the second transition. The resulting correlation can be expressed in terms of Legendre polynomials, with coefficients that depend on the spin sequence of the cascade and on the multipolarities of the emitted radiation.

In the present experiment, we study the ^{60}Co decay cascade, which proceeds predominantly through a 4–2–0 sequence and is therefore expected to exhibit a pronounced anisotropic angular correlation. The aim of the measurement is to determine the angular dependence of the coincidence rate, extract the corresponding angular-correlation coefficients, and compare the result with the theoretical expectation. At the same time, the experiment illustrates the practical requirements of precision coincidence spectroscopy, including the calibration of detector gains, discriminator thresholds, and timing delays, as well as the treatment of random coincidences, finite solid-angle effects, and residual geometric misalignment.

To this end, we use two scintillation detectors in a fast–slow coincidence setup. One detector remains fixed, while the second detector is moved through a range of angular positions around the ^{60}Co source. From the measured coincidence rates, together with the corresponding single rates and dedicated background measurements, we construct the corrected angular-correlation curve and test its compatibility with the expected 4–2–0 cascade shape.

II. PRE-LAB CONSIDERATIONS

A. Which distances to choose?

For one detector, the count rate is proportional to the covered solid angle:

$$N_i \propto \Omega_i \tag{1}$$

In the far-field limit, $\Omega_i \propto d^{-2}$, so the singles rate scales as

$$N_i \propto \frac{1}{d^2} \tag{2}$$

For two detectors at the same distance d, the true coincidence rate is therefore

$$R^t(\theta) \propto \Omega_1 \Omega_2 \propto \frac{1}{d^4} \tag{3}$$

Moving the detectors closer to the source thus improves the counting statistics quadratically with distance.

The drawback is the finite detector solid angle. The measured coefficients are attenuated according to [1]

$$A_{kk}^{\mathrm{exp}} = Q_k^{(1)} Q_k^{(2)} A_{kk}^{\mathrm{theory}}, \qquad k = 2, 4 \tag{4}$$

so the corrected coefficients are

$$A_{kk}^{\mathrm{theory}} = \frac{A_{kk}^{\mathrm{exp}}}{Q_k^{(1)} Q_k^{(2)}} \tag{5}$$

Here $Q_k^{(1)}$ and $Q_k^{(2)}$ are the correction factors for the respective detector, labeled by 1 and 2. If the statistical uncertainty of the measured coefficients scales with the coincidence statistics, then

$$\sigma\left(A_{kk}^{\mathrm{theory}}\right) \propto \frac{d^2}{Q_k^{(1)} Q_k^{(2)}} \tag{6}$$

The scaling $\sigma(A_{kk}^{\text{theory}}) \propto d^2/(Q_k^{(1)}Q_k^{(2)})$ follows from the combination of two effects: the statistical uncertainty grows as $1/\sqrt{N_{\text{coinc}}} \propto d^2$ because the coincidence rate decreases as d^{-4}. Larger coincidence count numbers therefore improve the precision of the extracted angular-correlation coefficients because the underlying counting statistics are Poissonian; therefore, the relative statistical uncertainty decreases as $1/\sqrt{N_{\text{coinc}}}$.

The optimal distance is not the largest possible one, where $Q_k \to 1$, where the statistics become poor, and also not the smallest possible one, where the solid-angle correction grows too strongly. Using the values provided in Siegbahn [2] , for $2'' \times 2''$ crystals, a source-to-detector distance of about 5 cm is the reasonable compromise. At 10 cm, the coincidence rate would already be smaller by a factor of 16, while the gain in correction factors is too modest to compensate for that loss. As a reference, the measured distance for the used detector setup of this experiment is 5.2(2) cm. Therefore, the measured source-to-detector distance from the experimental setup agrees with the reasonable compromise within its 1σ interval.

a. ***Experimental asymmetry.*** The asymmetry is not measured with the ideal point-detector coefficients, but with the attenuated coefficients A_{22}^{exp} and A_{44}^{exp}. Hence the describing function [3]

$$W_{\text{exp}}(\theta) = 1 + A_{22}^{\text{exp}} P_2(\cos\theta) + A_{44}^{\text{exp}} P_4(\cos\theta) \tag{7}$$

and the measured asymmetry becomes

$$\mathcal{A}_{\text{exp}} = \frac{W_{\text{exp}}(180°) - W_{\text{exp}}(90°)}{W_{\text{exp}}(90°)} \tag{8}$$

Finite solid angle can only suppress the anisotropy. Therefore, the measured asymmetry must be smaller than the ideal theoretical value before correction.

B. Which angles to pick?

For the 4–2–0 cascade, the angular correlation can be written as [1]

$$f(\theta) = A\left(1 + B\cos^2\theta + C\cos^4\theta\right) \tag{9}$$

With

$$\alpha = B + C, \qquad \beta = B - C \tag{10}$$

this becomes

$$f(\theta) = A\left[1 + \frac{\alpha+\beta}{2}\cos^2\theta + \frac{\alpha-\beta}{2}\cos^4\theta\right] \tag{11}$$

The point of this reparameterization is that one can directly inspect which angles constrain α and β most strongly.

Figures 1 and 2 show the results for the nominal values $\alpha = 0.1666$ and $\beta = 0.0830$. Varying α changes mainly the contrast between 180° and the side angles, so

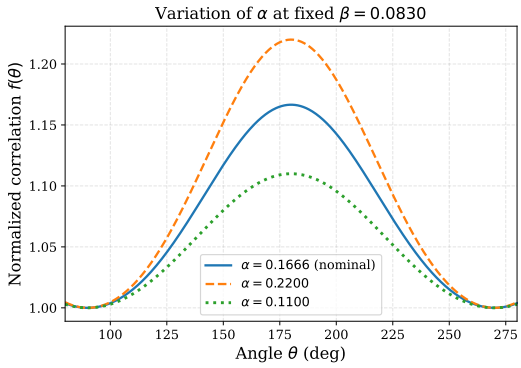


FIG. 1: Variation of the angular-correlation curve under changes of α at fixed β. The largest change occurs near 180°, while the curves meet near 90°.

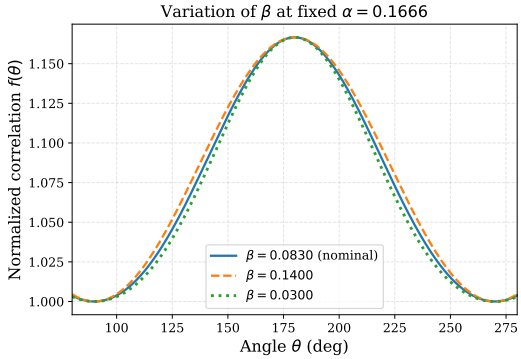


FIG. 2: Variation of the angular-correlation curve under changes of β at fixed α. The strongest effect appears around 135° and 225°.

α is best constrained near 180°. Varying β leaves the endpoints much less affected but changes the curvature strongly near 135° and 225°. At 90°, the sensitivity to both coefficients vanishes because $\cos\theta = 0$.

The most useful angles are, therefore:

- 90° as a normalization reference,
- 135° and 225° to constrain β,
- 180° to constrain α.

To cover these points, which provide the highest information density as well as enough intermediate points to gather statistics for assessing the goodness-of-fit, we choose a measurement range from 90.0° to 270.0° in steps of 15.0°. We note that the software didn't list explicit errors in the angles. However, we estimate the uncertainty to be on the last shown digit of the angles, here 0.1°, which likely is negligible in the total uncertainty estimation for the fit, when compared to the other sources of uncertainty.

C. How to correct for de-adjustment?

If the source is not exactly at the center of the detector circle, the source-to-detector distance of the movable detector changes with angle. Even if the true angular correlation were absent, this would generate a spurious angular modulation in the measured coincidence rate.

Under the simplified assumptions of no angular correlation, namely $B = C = 0$ from Equation (9) and no random coincidences, the singles rates are

$$f_{\mathrm{mov}}(\theta) \propto \frac{1}{d^2_{\mathrm{mov}}(\theta)}, \qquad f_{\mathrm{fix}}(\theta) \propto \frac{1}{d^2_{\mathrm{fix}}}. \tag{12}$$

Since the fixed detector is not moved, d_{fix} is constant from run to run, whereas $d_{\mathrm{mov}}(\theta)$ varies with angle. The coincidence rate therefore behaves as

$$f_{\mathrm{coinc}}(\theta) \propto \frac{1}{d^2_{\mathrm{fix}} d^2_{\mathrm{mov}}(\theta)} \tag{13}$$

The misalignment effect is therefore multiplicative and lives entirely in the movable branch of the detector setup.

To correct for this misalignment, the simplest correction is to divide the coincidence rate by the movable-detector singles rate:

$$\frac{f_{\mathrm{coinc}}(\theta)}{f_{\mathrm{mov}}(\theta)} \propto \frac{1}{d^2_{\mathrm{fix}}} = \text{const.} \tag{14}$$

Thus, the angle-dependent distance factor cancels. In practice, this means that the single rates are not just auxiliary counters; they are required to remove geometric imperfections of the setup. If the fixed branch also shows a weak modulation, one can generalize this to a correction proportional to $f_{\mathrm{fix}}/f_{\mathrm{mov}}$.

For the conducted experiment it was found, that both fixed and moving detector exhibited misalignment, which is why the finally used correction factor is $f_{\mathrm{fix}}/f_{\mathrm{mov}}$.

III. ANALYSIS TOOLS AND PSEUDO-DATA CHECK

A. Threshold scan

Figure 3 contains an immediate inconsistency. At threshold 0, the *source removed* dataset gives 5.11×10^6 counts, whereas the *source installed* dataset gives only 1.65×10^6 counts. This is unphysical as no source would naturally yield lower counts. Once the threshold is raised to 20, the source-removed rate collapses to only 600 counts, so the threshold-zero point is not a small fluctuation but a qualitatively different regime.

1. *Spectrum*

We fitted the peak region with

$$S(x) = G_1(x) + G_2(x) + b_0 + b_1 x, \tag{15}$$

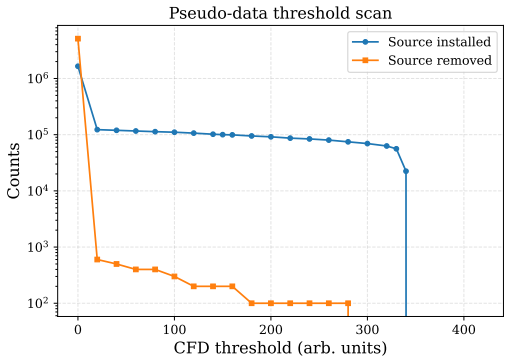


FIG. 3: Threshold scan of the CFD with and without source. The threshold-zero point is inconsistent with the rest of the scan.

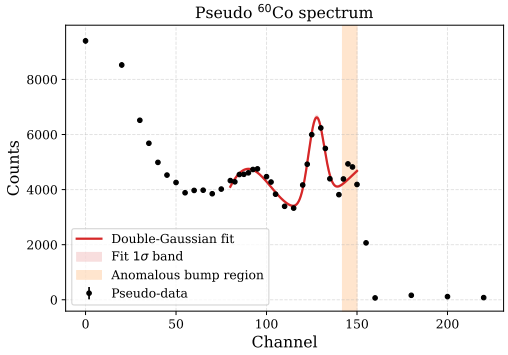


FIG. 4: Pseudo ^{60}Co spectrum with a double-Gaussian fit in the photopeak region. The red shaded region indicates the propagated 1σ covariance band of the fit, while the orange shaded high-channel bump is not part of the classical ^{60}Co decay spectrum which consists of two photopeaks [1] instead of 3.

that is, two Gaussian peaks on top of a linear background, where

$$G_i(x) = A \exp\left[-\frac{(x-\mu_i)^2}{2\sigma_i^2}\right] \tag{16}$$

The fit gives centroids

$$\mu_1 = 85.11(97), \qquad \mu_2 = 127.523(87), \tag{17}$$

but also

$$\chi^2/\mathrm{ndf} = 242.47/15. \tag{18}$$

The large χ^2 is caused by the extra bump between roughly channels 142 and 150, highlighted in Figure 4 . A sole ^{60}Co spectrum should not show such an additional structure in that region as the ^{60}Co spectrum only contains two photopeaks (see Figure 2 in [1]).

2. *Prompt curve*

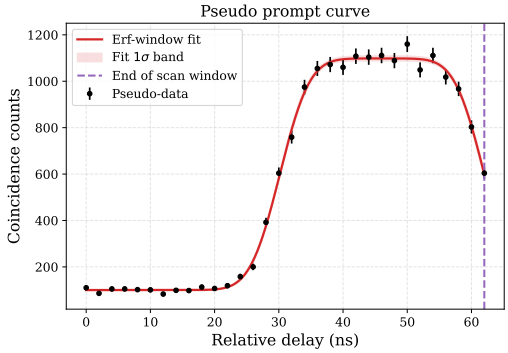


FIG. 5: Pseudo prompt curve fitted with the same erf-window model used in the main analysis. The shaded region indicates the propagated 1σ covariance band. The fitted right edge is driven to the last available scan point, which means the resolving time is not actually determined by the data.

We fitted the prompt curve with the same model that was later used in the main analysis,

$$P(t) = \frac{A}{2}\left[1 + \mathrm{erf}\left(\frac{t-(t_0 - w/2)}{\sigma_t}\right)\mathrm{erf}\left(\frac{(t_0 + w/2) - t}{\sigma_t}\right)\right] + A_0 \tag{19}$$

The fit returns

$$A = 998(12), \tag{20}$$
$$t_0 = 46.08(11)\,\mathrm{ns}, \tag{21}$$
$$w = 31.8(3)\,\mathrm{ns}, \tag{22}$$
$$\sigma_t = 5.2(2)\,\mathrm{ns}, \tag{23}$$
$$A_0 = 100(3) \tag{24}$$

with

$$\chi^2/\mathrm{ndf} = 25.87/27. \tag{25}$$

The numerical fit quality is statistically consistent with an p-value of $p = 53.68\%$, but that is misleading here. The decisive point is that the fitted right edge,

$$t_0 + \frac{w}{2} \approx 62.00\,\mathrm{ns}, \tag{26}$$

is effectively pinned to the last scanned value, 62 ns. The falling edge is therefore not actually observed all the way back to the baseline. The fit can still interpolate the visible points, but the resolving time w is not reliably determined because the scan stops too early.

3. *Angular correlation*

We fitted the pseudo angular-correlation data with

$$W(\theta) = A\left(1 + B\cos^2\theta + C\cos^4\theta\right) \tag{27}$$

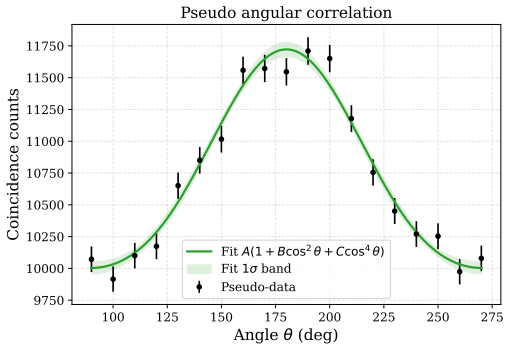


FIG. 6: Pseudo angular-correlation data fitted with the cosine-power parameterization of Equation (9) . The green shaded region indicates the propagated 1σ covariance band of the fit. The fitted shape is not fully consistent with the expected ^{60}Co values evident by scattering around the peak region at 180°.

which yields

$$A = 10\,000(50), \tag{28}$$
$$B = 0.09(3), \tag{29}$$
$$C = 0.08(3), \tag{30}$$

with

$$\chi^2/\mathrm{ndf} = 18.32/16. \tag{31}$$

The fit itself is statistically consistent with a p-value of $p = 30.55\%$. The problem is physical consistency. For the ideal 4–2–0 cascade one expects, in the same parameterization [1],

$$B_{\mathrm{th}} \approx 0.1248, \qquad C_{\mathrm{th}} \approx 0.0418. \tag{32}$$

The fitted C value is therefore almost a factor of two too large, and the fit value for B deviates roughly 33% from the theoretical expectation.

IV. ABBREVIATIONS

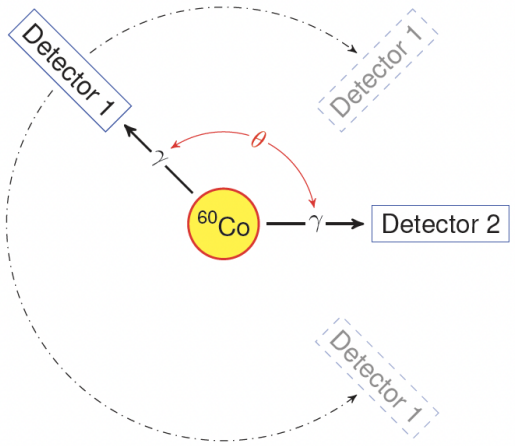
In the remainder of this report, we use several abbreviations in text or images, which we list below. For further details on how each component works internally, we refer to [4] .

- CFD = constant fraction discriminator

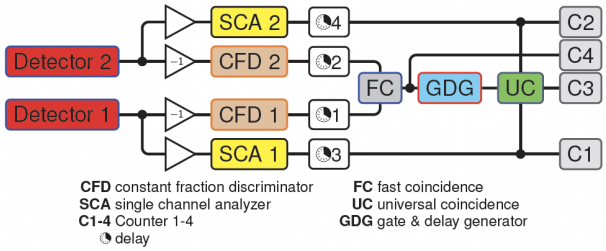
[1] These theoretical expectations are based on relating the correlation coefficients A_{22} and A_{44} of the Legendre-Polynomial expansion (see Equation (65)) to the A, B, C parametrization of Equation (9) . For concrete calculations, the values for A_{22} and A_{44} were used from [3] .

- FC = fast coincidence
- GDG = gate and delay generator
- MCA = multi-channel analyzer
- SCA = single-channel analyzer
- UC = universal coincidence

V. FAST-SLOW CIRCUIT CALIBRATION



(a) Geometric detector arrangement used for the angular-correlation measurement. Detector 2 remains fixed, while Detector 1 is moved on a circular path around the ^{60}Co source in order to vary the opening angle θ.



(b) Electronic fast–slow coincidence chain used to process the detector signals. Each detector is connected to an SCA branch and a CFD branch, which are combined downstream in the fast and universal coincidence units.

FIG. 7: Geometric and electronic layout of the γ-γ coincidence setup. The upper panel shows the detector arrangement around the ^{60}Co source, while the lower panel shows the full electronic processing chain with the two nominally identical fast and slow branches. Images adapted from [1] .

Before recording the angular-correlation data, we calibrated the complete electronic timing chain of the setup shown in Figure 7 . The upper panel displays the geometric arrangement of the two detectors around the ^{60}Co source, while the lower panel shows the full fast–slow coincidence electronics. Since the setup contains two detectors and therefore two nominally identical signal-processing branches, we carried out each calibration step separately for both sides. In particular, we adjusted the gain, discriminator settings, timing delays, and logic alignment for each detector branch before combining both channels at the coincidence stage.

This calibration procedure comprised the adjustment of the fast and slow branches and the synchronization of both at the coincidence stage. In the fast branch, we optimized the CFD thresholds and relative delays in order to maximize the true coincidence rate and determine the prompt timing response. In the slow branch, we aligned the logic outputs of the SCAs with the signal from the fast coincidence so that all inputs arrived simultaneously at the three-fold coincidence module. In the following, we describe each calibration step in more detail.

A. Main amplifier gains

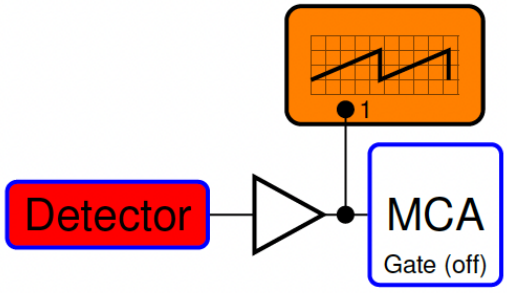


FIG. 8: Circuit to set and measure the amplitude gain using an oscilloscope. Each detector's respective signal is sent through the amplifier and monitored on an oscilloscope (shown in orange) to set the photopeak signal to an amplitude between 8 V and 10 V. Image taken from [1] .

We first adjusted the gain of the main amplifiers using the circuit shown in Figure 8 to maximize the signal amplitude while avoiding saturation of the downstream electronics. Using the oscilloscope in combination with the MCA, we monitored the photopeak pulses and tuned the amplifier output to lie in the range of 8 V to 9 V. This choice preserved a small safety margin to the maximum signal amplitude of 10 V. In the properly adjusted regime, the pulses exhibited a smooth, well-defined shape with a clean maximum and an undistorted decay. By contrast, saturation would have appeared as a clipping of the pulse peak. We distinguished the photopeak pulses from pure noise by varying the trigger level of the oscilloscope and selecting the region in which the waveform density



FIG. 9: Output signal of the main amplifier after setting the gain. The signal amplitude is in the range of 8 V to 9 V. Image taken from [5] .

was highest and the pulse shapes remained most stable. An example of the amplified output signals is shown in Figure 9 .

B. SCA windows

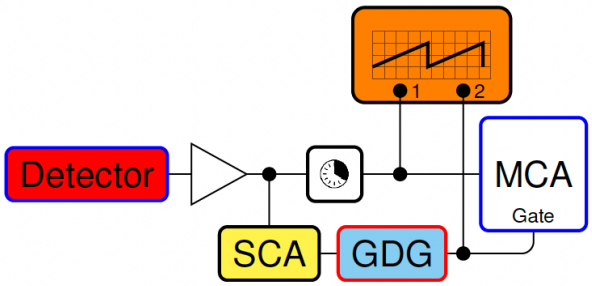
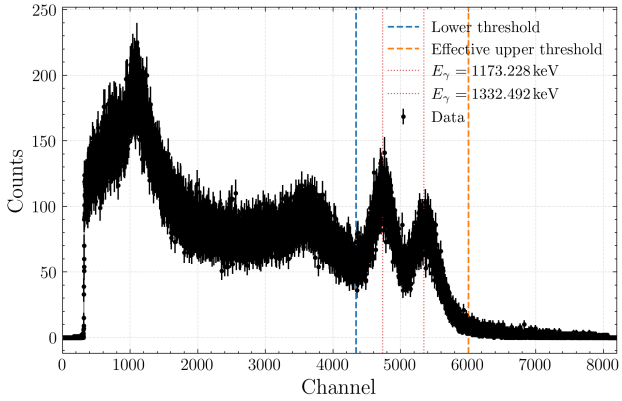
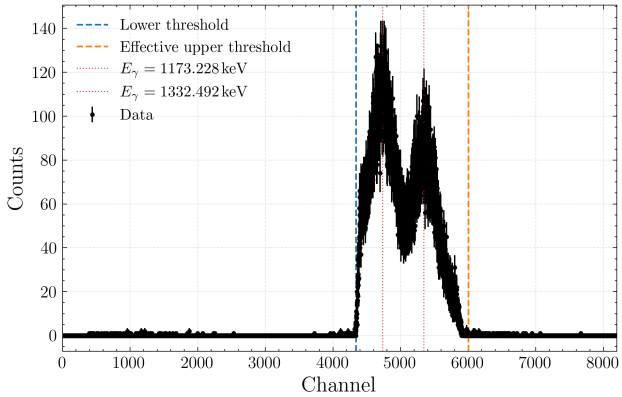


FIG. 10: Circuit used to set the SCA thresholds and record the corresponding ^{60}Co spectra with the MCA. The detector signal is sent through the calibrated main amplifier and split into an SCA–GDG branch and a delayed analogue branch, indicated by the clock symbol. The delayed analogue pulse and the SCA logic signal are monitored simultaneously with an oscilloscope, while the MCA records spectra with SCA and internal triggering. Image adapted from [1].

We then calibrated the SCAs, using the circuit shown in Figure 10 , to select pulses associated with the two ^{60}Co photopeaks while rejecting the remaining spectrum as efficiently as possible. For this purpose, we operated the MCA in coincidence mode and adjusted the SCA timing such that the photopeak signals were centered within the accepted logic window. We chose a common SCA window that covers both ^{60}Co lines rather than separating the two transitions. This choice increases the accepted count rate and therefore improves the coincidence statistics, since both detectors can register both gamma lines. At the same time, the sensitivity plateau remains close to $\eta = 1$ over the accepted region, so this configuration does not indicate a strong additional bias from the window setting itself.



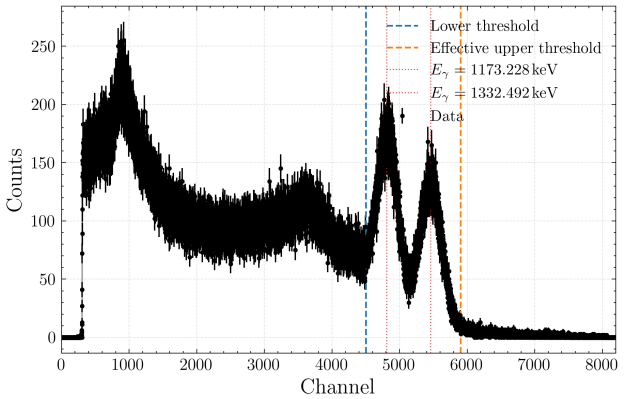
(a) Internally triggered spectrum of the movable detector.



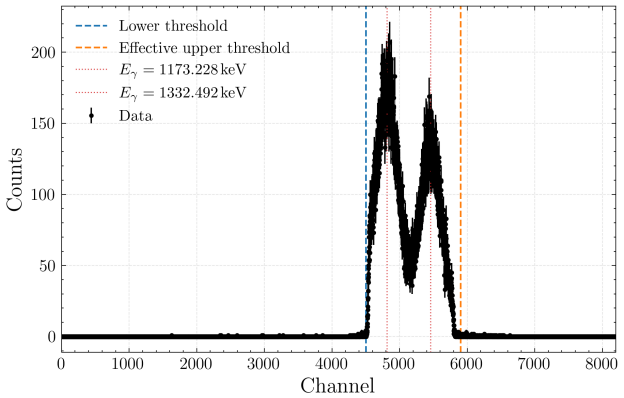
(b) SCA-triggered spectrum of the movable detector.

FIG. 11: Spectra of the movable detector recorded with internal and SCA triggering. The lower threshold marks the onset of the accepted SCA window, while the effective upper threshold indicates the channel at which the dominant accepted signal has already dropped off. The two dotted red lines indicate the fitted centroid positions of the 1173.228 keV and 1332.492 keV photopeaks.

Figures 11 and 12 compare the internally triggered and SCA-triggered spectra for the movable and fixed detectors. In both detectors, the SCA-gated spectra retain the two photopeaks while suppressing events outside the selected window. We determined the lower threshold from the sensitivity curves, shown in Figures 13a and 13b , as the onset at which the gated spectrum rises from essentially zero to the flat acceptance plateau. We defined the effective upper threshold from the channel at which the bulk of the accepted signal drops rapidly and the sensitivity falls below approximately $\eta \approx 0.1$. This criterion yields effective upper thresholds near channel 6000 for



(a) Internally triggered spectrum of the fixed detector.



(b) SCA-triggered spectrum of the fixed detector.

FIG. 12: Spectra of the fixed detector recorded with internal and SCA triggering. As for the movable detector, the lower threshold marks the onset of the accepted region, while the effective upper threshold indicates the channel above which the dominant accepted signal has already dropped off strongly. The two dotted red lines indicate the fitted centroid positions of the 1173.228 keV and 1332.492 keV photopeaks.

the movable detector and near channel 5900 for the fixed detector. Above these values, both detectors still exhibit a weak residual tail with non-zero counts. One explanation is that the SCA window was set slightly too wide, such that a small fraction of higher-amplitude pulses still passed the logic selection. However, the counts are two orders of magnitude smaller and therefore likely won't have significant impact in the further measurements.

To localize the two photopeaks more precisely, we fitted each peak with a Gaussian (dependent on the channel number ch) plus constant background,

$$N(ch) = A \exp\left[-\frac{(ch-\mu)^2}{2\sigma^2}\right] + C \tag{33}$$

where A denotes the peak amplitude, μ the centroid channel, σ the peak width, and C a constant offset. We use the fitted centroids only to indicate the positions of the ^{60}Co lines in the spectra and sensitivity plots; the threshold determination itself is based on the measured acceptance behaviour. The obtained fit parameters are not explicitly listed, as merely the mean is used to accurately indicate the peak position in the plotted spectra.

1. *SCA sensitivity*

To quantify the SCA acceptance, we formed the sensitivity [1]

$$\eta(ch) = \frac{R_{\mathrm{gated}}(ch)}{R_{\mathrm{self}}(ch)} \tag{34}$$

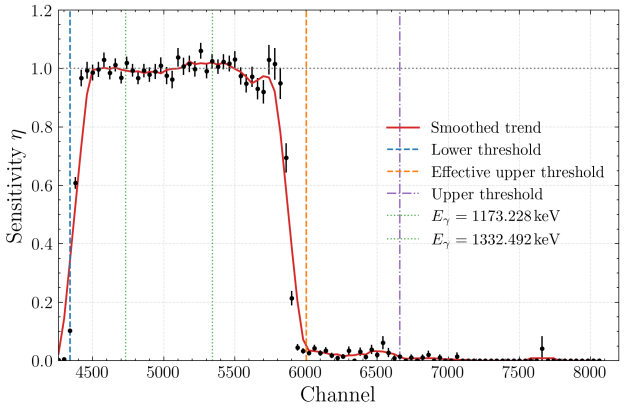
that is, the ratio of the SCA-gated count rate to the internally gated count rate in each rebinned channel. [2] The uncertainty was propagated from the Poisson counting statistics of both spectra.

No measurement uncertainty for the time was listed by the software, however, we estimate the influence of the time uncertainty to be negligible compared to the counting uncertainty coming from the Poisson distribution.

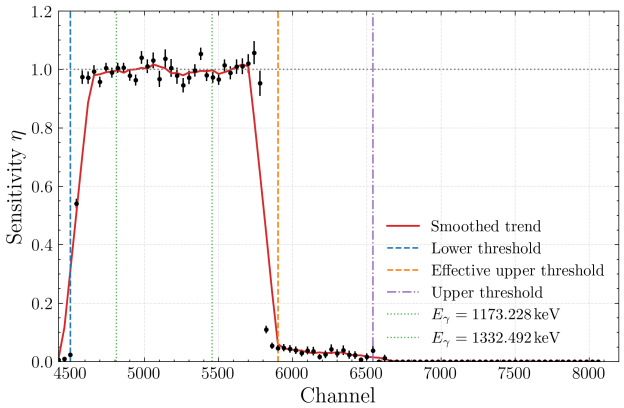
The resulting sensitivity curves are shown in Figure 13 . Both detectors exhibit the same qualitative behaviour. Below the lower threshold, the sensitivity is consistent with zero, as expected since the SCA rejects pulses in this region. At the lower threshold, η rises sharply to a plateau near unity. Across most of the accepted window, the sensitivity fluctuates weakly around $\eta = 1$, which indicates that the gated and internally triggered spectra agree in count rate once the pulse height lies inside the selected region. This behaviour confirms that the SCA accepts the photopeak region with high efficiency. At the high-channel edge, the sensitivity drops rapidly near channel 6000 for the movable detector and near channel 5900 for the fixed detector, which motivates our definition of the effective upper thresholds (which is solely based on visual inspection). Beyond this edge, both curves retain a small non-zero tail. As in the raw spectra, this residual acceptance might reflect a slightly too wide SCA window or a small leakage of larger-amplitude pulses into the accepted region.

We note that the red curve in Figure 13 is not an independent fit. We obtained it from a five-point moving average, in which each smoothed value is assigned to the central point of the corresponding five-bin window. This is done to guide the eye and make the window boundaries more visible. The nearly flat top part of this smoothed curve, again centered around $\eta = 1$, confirms that the SCA transmits the photopeak region with effectively the same rate as the internally triggered spectrum while suppressing pulses outside the selected range.

[2] This is equivalent to the ratio of the individual counts as both measurement were conducted for the same time of 30 s.



(a) Sensitivity curve of the movable detector.



(b) Sensitivity curve of the fixed detector.

FIG. 13: SCA sensitivity curves of the movable and fixed detectors. The black points give the measured sensitivity η, the red line shows a moving average over five consecutive rebinned points, and the vertical lines indicate the lower threshold, the effective upper threshold, and the fitted centroid positions of the two ^{60}Co photopeaks. In the SCA sensitivity plots, the additionally shown upper-threshold line marks include the residual counts of higher channels, while the effective upper threshold indicates the position at which the dominant accepted signal has already dropped off strongly.

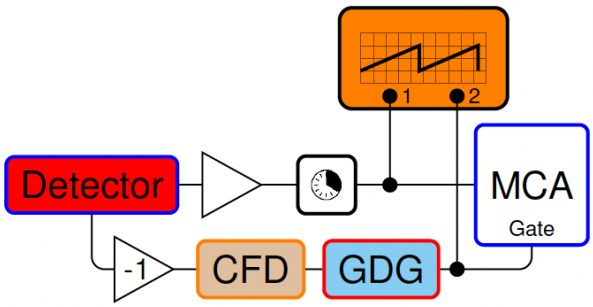


FIG. 14: Circuit used to adjust the CFD thresholds and to record the corresponding ^{60}Co spectra. The detector signal is split into the analogue branch sent to the MCA and the CFD branch that generates the logic gate. By comparing internally triggered and CFD-triggered spectra, one can determine whether the chosen threshold suppresses noise while retaining the photopeak region. Image adapted from [1].

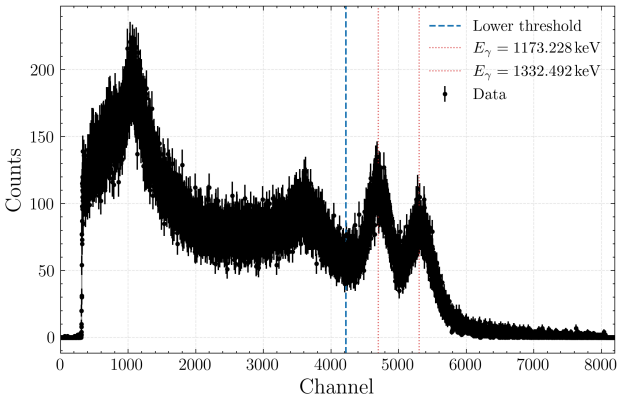


FIG. 15: Oscilloscope signal used to verify the CFD adjustment. The discriminator threshold must be high enough to suppress electronic noise, but low enough that genuine detector pulses still produce stable logic signals. Image adapted from [5] .

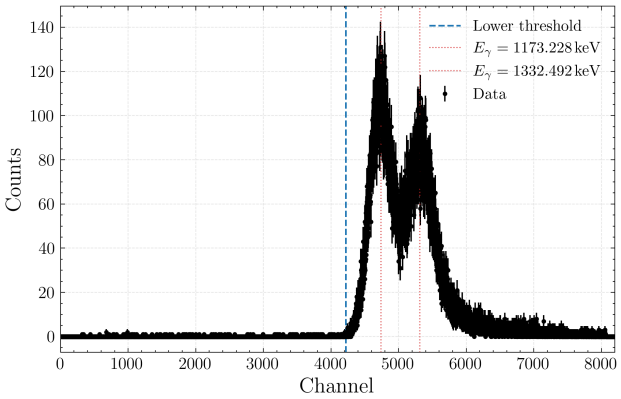
C. CFD thresholds

Next, we optimized the CFDs by choosing thresholds that suppress electronic noise without removing genuine detector pulses. This adjustment is critical because a threshold set too low allows the discriminator to trigger on noise, whereas a threshold set too high rejects valid events with smaller amplitudes. We therefore recorded spectra with internal gating and with CFD-generated gating and selected thresholds that retained the photopeak region while reducing the low-amplitude background as much as possible. In contrast to the SCA discussed in Section V B , the CFD allows us to define only a lower threshold. It therefore removes the low-amplitude noise contribution but does not impose any independent upper cut on large-amplitude pulses.

Figures 16 and 17 compare the internally triggered and CFD-triggered spectra for the movable and fixed detectors. In both cases, the CFD suppresses the low-channel region, which demonstrates that the threshold removes the major part of the electronic noise. At the same time, the two ^{60}Co photopeaks remain clearly visible, which shows that genuine pulses are still accepted reliably. Unlike the SCA spectra, however, the CFD-triggered spectra retain a number of events at higher channel values. This behaviour follows directly from the fact that the CFD provides only a lower threshold and therefore does not reject pulses with larger amplitudes once they exceed the discriminator setting. The threshold shown as blue dashed vertical lines in Figures 16 and 17 were again ob-

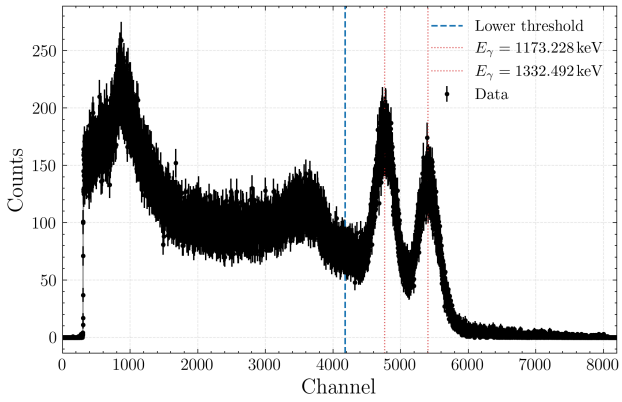


(a) Internally triggered spectrum of the movable detector.

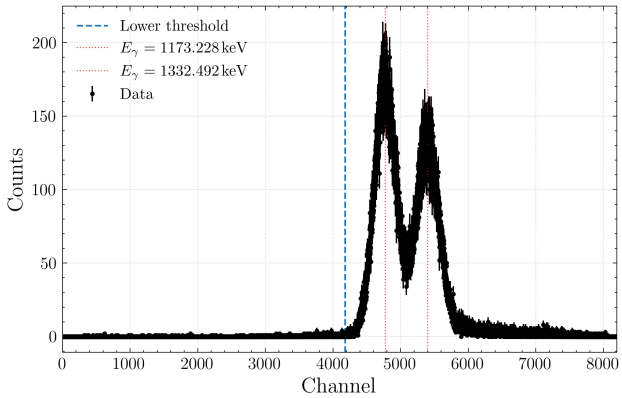


(b) CFD-triggered spectrum of the movable detector.

FIG. 16: Spectra of the movable detector recorded with internal and CFD triggering. The lower threshold marks the onset above which the CFD begins to accept detector pulses. Since the CFD imposes only a lower threshold, the high-channel part of the spectrum remains populated and is not cut off in the same way as for the SCA spectra. The two dotted red lines indicate the photopeak positions of the 1173.228 keV and 1332.492 keV lines, determined as described in Section V B .



(a) Internally triggered spectrum of the fixed detector.



(b) CFD-triggered spectrum of the fixed detector.

FIG. 17: Spectra of the fixed detector recorded with internal and CFD triggering. As for the movable detector, the CFD threshold suppresses the low-amplitude part of the spectrum, while the high-channel region remains visible because no upper threshold is applied. The dotted red lines again indicate the fitted centroid positions of the two ^{60}Co photopeaks, obtained as described in Section V B.

tained by inspection of the sensitivity curves shown in Figure 18 .

1. *CFD sensitivity*

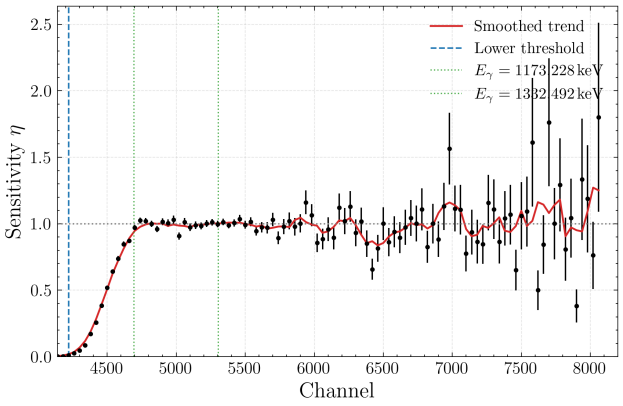
To quantify the acceptance of the CFD threshold, we again formed the sensitivity [1]

$$\eta(ch) = \frac{R_{\mathrm{gated}}(ch)}{R_{\mathrm{self}}(ch)}, \tag{35}$$

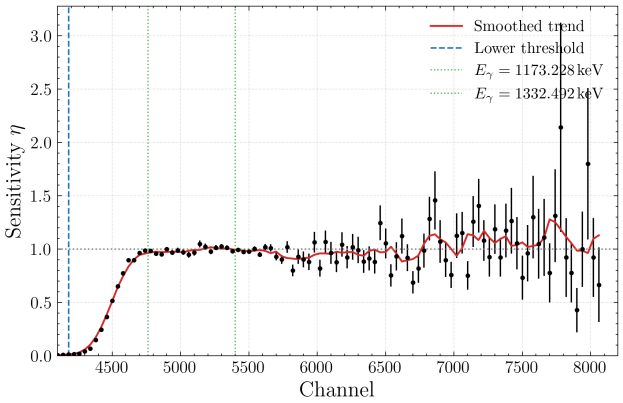
that is, the ratio of the CFD-triggered count rate to the internally triggered count rate in each rebinned channel. Again, the ratio of both raw counts is equivalent as the measurement time was 30 s in both cases. The measurement time uncertainty is neglected as described in Section V B .

The CFD sensitivity curves in Figure 18 differ from the SCA case. Since the CFD introduces only a lower threshold, the sensitivity rises sharply from zero at the threshold and approaches a plateau around $\eta \approx 1$, but it does not exhibit a sharp upper cutoff. Instead, the high-channel region remains populated and the corresponding sensitivity points become increasingly irregular. This behaviour is expected, because the CFD does not exclude large-amplitude pulses once they are above threshold.

The pronounced fluctuations at large channel numbers are not physically meaningful. In this regime, both the internally triggered and CFD-triggered spectra contain only a small number of counts in each bin, so the ratio becomes dominated by statistical fluctuations. This in-



(a) Sensitivity curve of the movable detector.



(b) Sensitivity curve of the fixed detector.

FIG. 18: CFD sensitivity curves of the movable and fixed detectors. The black points give the measured sensitivity η, while the red line shows the same five-point moving average that was already introduced for the SCA sensitivity in Section V B 1 . The lower threshold marks the onset above which the CFD accepts the detector pulses. At high channel values the curves become irregular, but these fluctuations are not physically meaningful because the counting statistics are poor there, as indicated by the large error bars.

terpretation is supported directly by the large error bars in the high-channel region. In addition, visual inspection of the spectra shown in Figures 17 and 16 shows that the count rate for channels above the photopeak ones (roughly larger than 6000 channels) is two orders of magnitude smaller than the one of the photopeaks, indicating relative unimportance in terms of raw counts.

Near the two photopeaks, by contrast, both detectors show the same behaviour as in the SCA case: the sensitivity fluctuates around $\eta = 1$, which indicates that the CFD accepts the photopeak pulses with effectively the same efficiency as the internally triggered spectrum.

As in Section V B 1 , the red curve is not an independent fit. We obtained it by applying a moving average over five consecutive rebinned points in order to guide the eye and make the threshold behaviour more visible. Around the photopeak region, this smoothed trend remains close to unity, whereas at high channel values it mainly illustrates the onset of the statistically unstable tail rather than a robust physical structure.

D. Fast branch delay

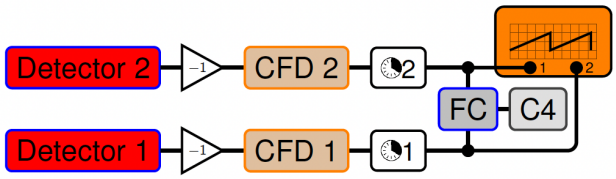


FIG. 19: Circuit used to calibrate the relative delay in the fast branch. The two detector signals are processed by their CFDs and sent to the fast-coincidence module. One delay is kept fixed, while the second is varied in discrete steps. For each setting, the fast-coincidence output is counted with Counter 4 in order to determine the prompt coincidence curve. Image adapted from [1] .

After establishing the energy selection, we aligned the fast timing branch by matching the arrival times of the two CFD output signals at the fast-coincidence module. Using the circuit shown in Figure 19 , we kept one delay fixed, varied the second delay in discrete steps, and recorded the fast-coincidence counts for each setting. This scan allowed us to identify the delay that maximizes the true coincidence rate and to determine the width of the prompt coincidence window. The fast-branch calibration therefore provides both the optimal timing condition for the main measurement and the characteristic coincidence window relevant for the subsequent analysis of random coincidences.

1. *Prompt curve*

To model the prompt coincidence curve, we fitted the measured counts as a function of the relative delay t with [1]

$$f(t) = \frac{A}{2}\left[1 + \mathrm{erf}\left(\frac{t-(t_0 - w/2)}{\sigma_t}\right)\mathrm{erf}\left(\frac{(t_0 + w/2) - t}{\sigma_t}\right)\right] + A_0 \tag{36}$$

Here, A_0 denotes the constant baseline of accidental counts outside the prompt window, A is the plateau height above this baseline, t_0 is the center of the coincidence window, w is its width, and σ_t characterizes the steepness of the rising and falling edges. In physical terms, w corresponds to the effective resolving time of the

fast coincidence, while σ_t reflects the temporal smearing of the two edges due to the finite timing resolution of the electronics.

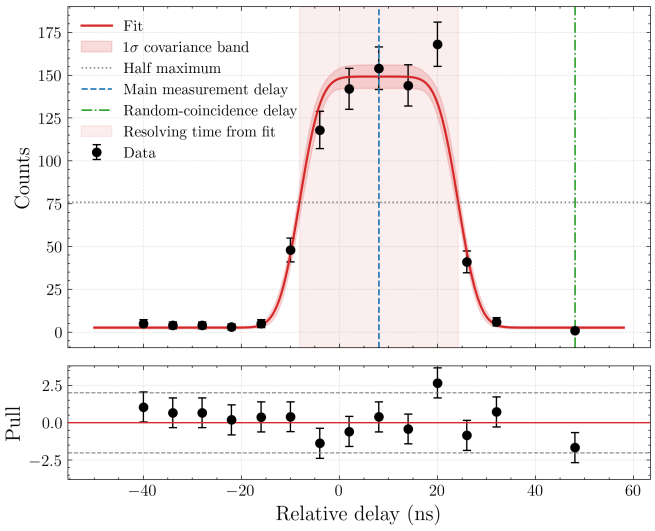


FIG. 20: Prompt coincidence curve obtained from the fast-branch delay scan. The black points show the measured coincidence counts, the red curve gives the fit, and the shaded band indicates the propagated 1σ covariance band of the fit parameters. The blue dashed line marks the delay chosen for the main measurement, the green dash-dotted line marks the delay used for the random-coincidence measurement discussed in Section VI A , and the red shaded vertical region indicates the resolving-time window $t_0 \pm w/2$ obtained from the fit.

The resulting fit is shown in Figure 20 . We obtained

$$t_0 = 8.0(3)\,\text{ns}, \tag{37}$$
$$w = 32.4(10)\,\text{ns}, \tag{38}$$
$$\sigma_t = 4.8(8)\,\text{ns}, \tag{39}$$
$$A_0 = 2.7(7)\ \text{counts}, \tag{40}$$
$$A = 147(7)\ \text{counts}, \tag{41}$$

with

$$\chi^2/\text{ndf} = 15.95/9 \tag{42}$$

The quoted count parameters follow directly from the fitted rates and the fixed measuring time of 30 s per scan point.

The χ^2 value is likely explained by the fact that the measurement point at 22 ns exhibits a pull of roughly 2.5σ, the one at 48 ns as well as at -4 ns a pull of roughly 2σ. In comparison, all other points agree with the 0 pull baseline within their measurement 1σ uncertainty. Therefore, this seems to be the likely cause of the deviation from unity of χ^2/ndf.

The fitted width w defines the resolving time of the fast coincidence. Generally, if the resolving time is chosen too short, the prompt curve becomes too narrow and true coincidences are lost, so the plateau can shrink and become rounded; if it is chosen too long, the plateau broadens in time and more random coincidences are accepted, even though the intrinsic edge width remains determined mainly by the timing resolution of the electronics.

In the present measurement, the fitted prompt window extends nominally from approximately

$$t_0 - \frac{w}{2} \approx -8.1\,\text{ns} \qquad \text{to} \qquad t_0 + \frac{w}{2} \approx 24.2\,\text{ns}, \tag{43}$$

so that the delay chosen for the main measurement,

$$t_{\text{main}} = 8\,\text{ns}, \tag{44}$$

lies inside the prompt plateau and close to its center. By contrast, the delay chosen for the random-coincidence measurement,

$$t_{\text{rand}} = 48\,\text{ns}, \tag{45}$$

lies far enough outside the prompt window, where the coincidence rate has already dropped back to the accidental baseline.

We therefore conclude that the chosen resolving time is sufficient for the main measurement.

E. Slow branch delay

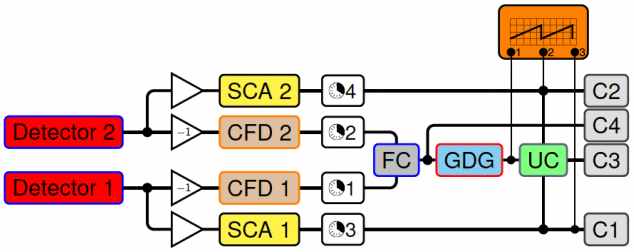


FIG. 21: Circuit used to synchronize the slow branch with the fast coincidence branch at the three-fold coincidence stage. The two SCA outputs and the delayed fast-coincidence signal are brought together at the universal coincidence unit. The SCA delays and the GDG timing are adjusted such that all logic pulses overlap in time and produce a stable three-fold coincidence. Image adapted from [1] .

Finally, we synchronized the slow branch with the fast branch at the three-fold coincidence stage. To achieve this alignment, we adjusted the SCA delays and, where necessary, the gate-and-delay-generator timing such that the SCA outputs and the fast-coincidence signal overlapped in time at the coincidence unit. We also matched the widths of the logic pulses as closely as visually possible in order to ensure reliable coincidence formation.

This procedure established the final timing configuration used for the measurement of the angular correlation.



FIG. 22: Oscilloscope traces of the properly aligned fast and slow logic pulses. The displayed signals correspond to the two SCA outputs and the delayed fast-coincidence signal at the input of the universal coincidence unit. Their temporal overlap ensures that a true event can generate the required three-fold coincidence. Image adapted from [5] .

Figure 21 shows the relevant part of the electronic chain. The two SCA outputs from the slow branch and the delayed fast-coincidence signal serve as the three inputs to the UC module. In order to register a true event reliably, the three logic signals must overlap sufficiently in time. We therefore adjusted the delay modules in the two SCA branches and, if necessary, the GDG delay in the fast branch until all three pulses arrived simultaneously at the coincidence unit. The aligned signals are shown in the oscilloscope screenshot of Figure 22 . Here, the yellow signal corresponds to the GDG output and the purple and blue are the slow-branch signals. In practice, the relevant criterion is not the exact coincidence of one particular edge alone, but the temporal overlap of the full logic pulses. Nevertheless, aligning the centers of the pulses provides the most robust operating condition, because it maximizes the common overlap interval and therefore the probability that true coincidences are accepted even in the presence of small timing fluctuations. At the same time, the pulse widths should be chosen as similarly as possible, so that no branch dominates the coincidence condition by being significantly broader than the others. For the oscilloscope triggering, we found that using the output of the fast coincidence was more convenient than triggering on one of the SCA outputs. The fast-coincidence signal is more sharply localized in time and therefore provides a more stable temporal reference on the screen. By contrast, triggering on one SCA output leads to a less well-defined visualization of the relative pulse positions, since the slow-branch signals are broader and less localized. Using the fast-coincidence output as the trigger thus made it easier to judge whether the two SCA signals were centered on the delayed fast signal and whether the pulse overlap was sufficient for stable operation.

VI. MEASUREMENT OF THE ANGULAR γ-γ CORRELATION

After completing the calibration of the detector gains, discriminator thresholds, and timing delays, we recorded the angular-correlation data by varying the opening angle between the two detectors and measuring the corresponding coincidence rate. For all following discussed measurements, we scanned through opening angles ranging from 90.0° to 270.0° in steps of 15° as discussed in Section II B . For each angular setting, we also recorded the single rates of both detectors, which later allowed us to monitor stability, estimate the random-coincidence contribution, and correct for residual geometric misalignment of the setup. The measured coincidence data were then processed by subtracting the dedicated random-coincidence measurement angle by angle and by applying the singles-based misalignment correction (for a given angle) before performing the final angular-correlation fits.

A. Measurement of random coincidences

To quantify the accidental background of the coincidence setup, we performed a dedicated measurement of the random-coincidence rate. For this purpose, we shifted the relative delay in the fast branch outside the prompt coincidence window to 48 ns, as discussed in Section V D 1. In this configuration, true prompt coincidences are suppressed, while uncorrelated overlaps of signals from the two detector branches can still generate coincidences at a low rate. The resulting count rate therefore provides a direct experimental estimate of the random-coincidence contribution that is later subtracted from the main angular-correlation measurement.

For each angle, we recorded the coincidence counts in this displaced-delay configuration and combined repeated measurements by summing the total counts and total measuring times before forming the rate. This procedure is statistically preferable to averaging individual rates because it automatically assigns the correct Poisson weight to runs with different durations or count numbers. The corresponding uncertainty was obtained from the combined counting statistics according to

$$R_{\mathrm{rand}}(\theta) = \frac{N_{\mathrm{rand,tot}}(\theta)}{T_{\mathrm{tot}}(\theta)} \qquad \sigma_{R_{\mathrm{rand}}}(\theta) = \frac{\sqrt{N_{\mathrm{rand,tot}}(\theta)}}{T_{\mathrm{tot}}(\theta)} \tag{46}$$

In addition to this direct measurement, we compared the data with the standard expectation for accidental coincidences:

$$R_{\mathrm{rand,pred}}(\theta) = \Delta t_{\mathrm{res}}\, R_1(\theta)\, R_2(\theta) \tag{47}$$

where $\Delta t_{\mathrm{res}} = w$ is the resolving time obtained from the prompt-curve fit in Section V D 1, and R_1 and R_2 are the single rates of the two detectors. The uncertainty of the

predicted random-coincidence rate was propagated from the uncertainties of Δt_{res}, R_1, and R_2.

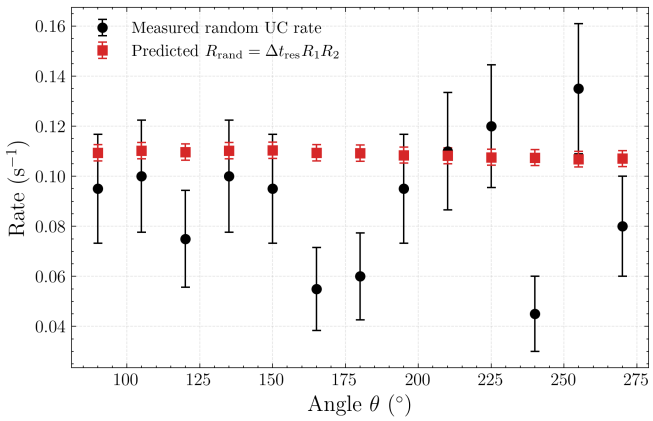


FIG. 23: Measured and predicted random-coincidence rates as a function of the detector angle. The black points show the directly measured random-coincidence rate in the displaced-delay configuration, while the red squares show the prediction $R_{\mathrm{rand}} = \Delta t_{\mathrm{res}} R_1 R_2$ based on the measured single rates and the resolving time Δtres from the prompt-curve fit. The error bars correspond to the propagated counting uncertainties of the measured and predicted rates, respectively.

Figure 23 compares the directly measured random-coincidence rate with the prompt-based prediction. The predicted rate is nearly independent of angle because the singles rates vary only weakly across the scan, so that the product $R_1 R_2$ remains approximately constant. The directly measured random-coincidence rate, by contrast, exhibits visibly larger scatter. This behaviour is expected, since the number of random coincidences is small and therefore subject to comparatively large Poisson fluctuations. The error bars of the measured points are correspondingly larger than those associated with the predicted values.

Although the direct random-coincidence data fluctuate from point to point, they remain of the same order of magnitude as the prediction from $\Delta t_{\mathrm{res}} R_1 R_2$ and partially agree within their respective 1σ error band. We nevertheless use the directly measured random-coincidence rate angle by angle in the final analysis, because it constitutes the more direct experimental estimate of the accidental background under the actual measurement conditions. This procedure is especially appropriate here because the same angles were measured both in the dedicated random-coincidence run and in the main angular-correlation scan, which allows the random contribution to be subtracted pointwise from the coincidence data.

B. Detector stability

To assess the stability of the setup over the full measurement period, we compared repeated measurements taken at the same angles throughout the scan. Since the angular-correlation measurement was performed by moving only one detector, the designation of the fixed detector requires some care in the interpretation of the plots. The fixed detector always remained at one physical angle in the laboratory frame, while the movable detector was rotated through the different angular settings. Therefore, when we plot the count rate of the fixed detector for different angles, these points do not indicate that the fixed detector itself was moved. Rather, they represent the count rates measured by the fixed detector in those runs in which the movable detector was set to the corresponding angle.

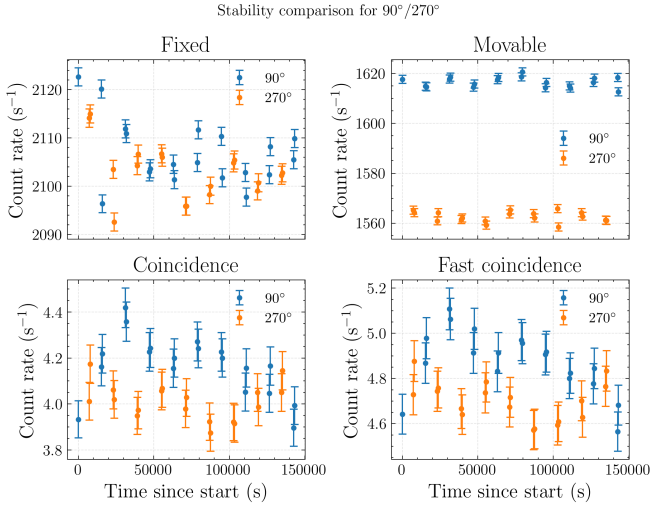


FIG. 24: Stability comparison for the repeated measurements at 90° and 270°. The four panels show the count rates of the fixed detector, the movable detector, the coincidence channel, and the fast-coincidence channel as a function of time since the start of the measurement campaign. The two angle settings correspond to equivalent geometries and therefore provide a direct consistency check of the setup stability.

To exploit the symmetry of the angular-correlation geometry, we compared measurements taken at scan settings that correspond to the same physical opening angle [3] , namely 90°/270°, 105°/255°, 120°/240°, 135°/225°, 150°/210°, and 165°/195°, together with the standalone 180° setting. These angle pairs satisfy $\phi' = 360° - \phi$ and

[3] Both detectors being in front of each other corresponds to an opening angle of 180°. Then, whether the angle increases clockwise or counterclockwise rotations is a mere choice of coordinates that won't alter the physics.

therefore represent equivalent detector geometries. In an ideally centered setup, they should yield identical count rates within statistical fluctuations. These repeated measurements were not merged here but were instead used as a direct test of the reproducibility and long-term stability of the setup.

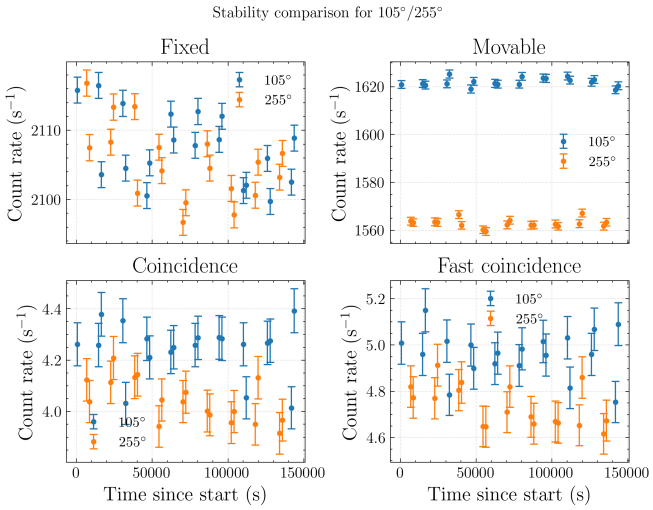


FIG. 25: Stability comparison for the repeated measurements at 105° and 255°. As in Figure 24, the four panels compare the fixed-detector, movable-detector, coincidence, and fast-coincidence count rates for a pair of geometrically equivalent angles.

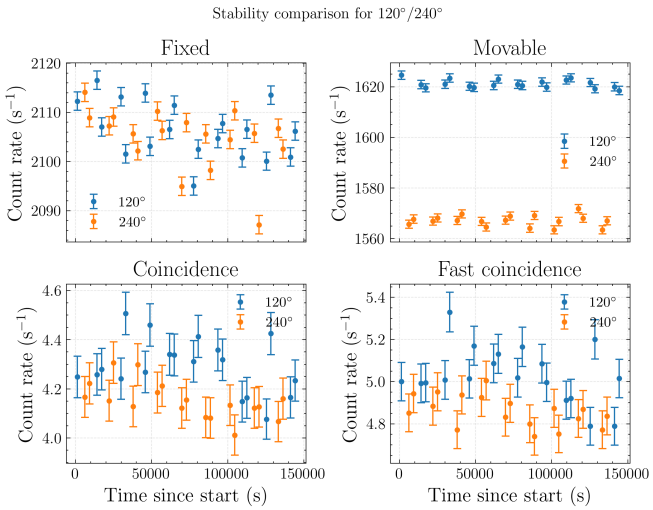


FIG. 26: Stability comparison for the repeated measurements at 120° and 240°. The paired measurements allow one to test whether the detector rates remain reproducible over time for two settings that should correspond to the same physical geometry.

Figures 24–30 show the time evolution of the singles and coincidence channels for all repeated angle pairs. The movable detector appears comparatively stable over

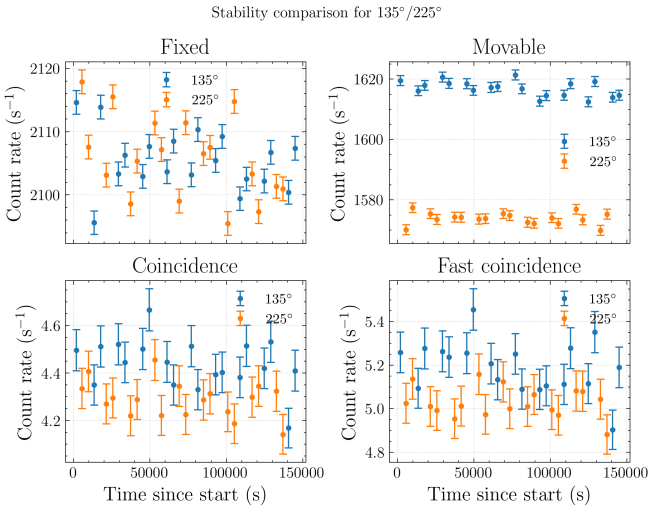


FIG. 27: Stability comparison for the repeated measurements at 135° and 225°. The four panels again compare the singles and coincidence channels for two equivalent angular settings distributed across the full data-taking period.

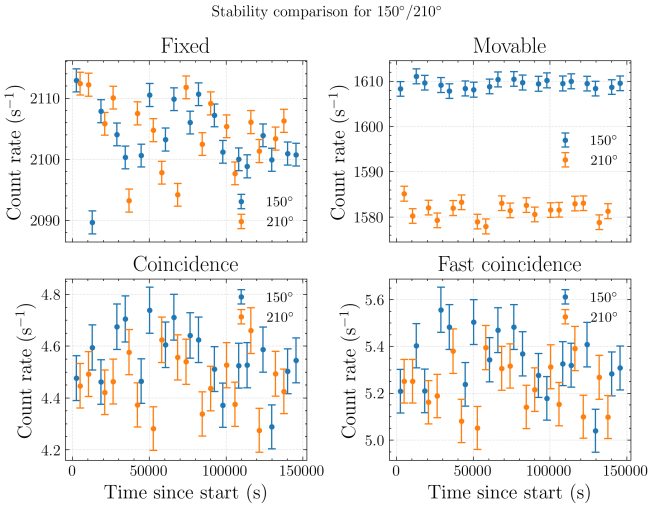


FIG. 28: Stability comparison for the repeated measurements at 150° and 210°. These repeated measurements show how the fixed-detector, movable-detector, coincidence, and fast-coincidence rates evolve over the course of the scan for another pair of geometrically equivalent angles.

time. Within each angle pair, the count rate remains clustered in a narrow band, and the repeated measurements at the two equivalent angles agree within their statistical uncertainties. However, comparing the count rates of the angle pairs shows clear deviations for every single angle pair, though the gap is less pronounced for the 165°-195° pair. The difference in count rates stays in the range of $10\,\mathrm{s}^{-1}$ to $60\,\mathrm{s}^{-1}$. This noticeable difference is likely a direct effect of the source 60Co being off-center,

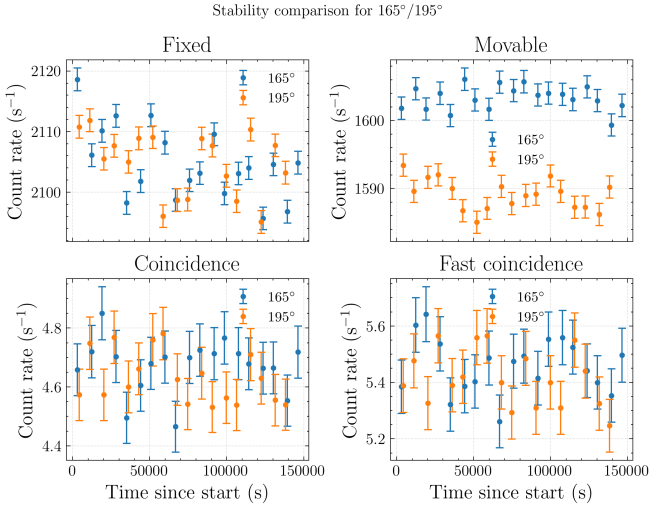


FIG. 29: Stability comparison for the repeated measurements at 165° and 195°. The comparison probes whether the measured rates remain reproducible at late stages of the scan for two nominally identical geometries.

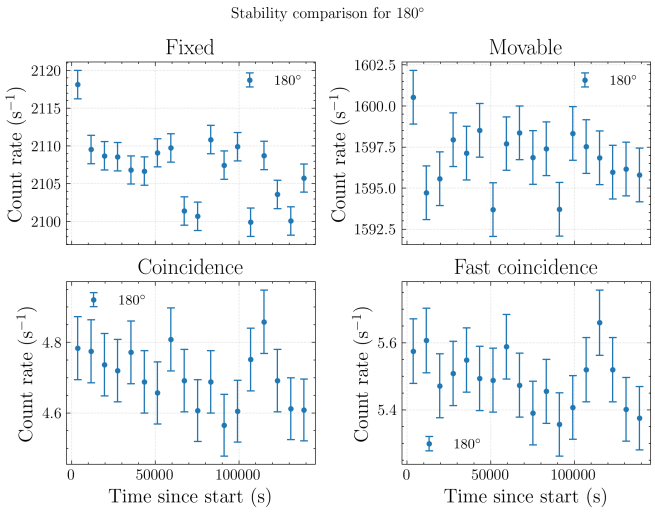


FIG. 30: Stability comparison for the repeated measurements at 180°. Since this angle maps onto itself under a 180° shift, only one set of repeated measurements exists. The four panels therefore show the temporal behaviour of the fixed-detector, movable-detector, coincidence, and fast-coincidence channels for the same nominal geometry revisited throughout the scan.

thereby introducing misalignment effects, which are further discussed in the upcoming Section VI C .

By contrast, the fixed detector exhibits visibly larger scatter. Although its absolute rate changes only at the level of a few s^{-1}, these fluctuations are larger than those seen for the movable detector and are also larger relative to the quoted statistical uncertainties. In this sense, the fixed detector is less stable than the movable detector over the duration of the full scan.

The coincidence and fast-coincidence channels fluctuate more strongly than the singles rates, as expected from their much lower count rates of 3 orders of magnitude less and correspondingly larger relative statistical uncertainties. The fast coincidence rate is also larger than the coincidence rate for all angles, which is expected as for every real coincidence a fast coincidence is required. In the present data, the repeated measurements show that the setup is not perfectly stable over the full measurement period. Linear as well as a constant model fit have been tried to fit to the data, resulting in goodness-of-fits values of $\chi^2/\mathrm{ndf} = 7$ to $\chi^2/\mathrm{ndf} = 10$, indicating no statistical compatibility between these models. As there is no clear trend visible to us in the data, we conclude that there was no clear drift for any of the detectors at any of the angles during the $\sim 40\,\mathrm{h}$ data-taking interval. The stability study therefore shows that the setup remained sufficiently reproducible for the angular-correlation measurement, while at the same time indicating that residual time-dependent fluctuations, especially in the fixed-detector channel, must be kept in mind when interpreting small deviations in the final angular distribution.

C. Misalignment

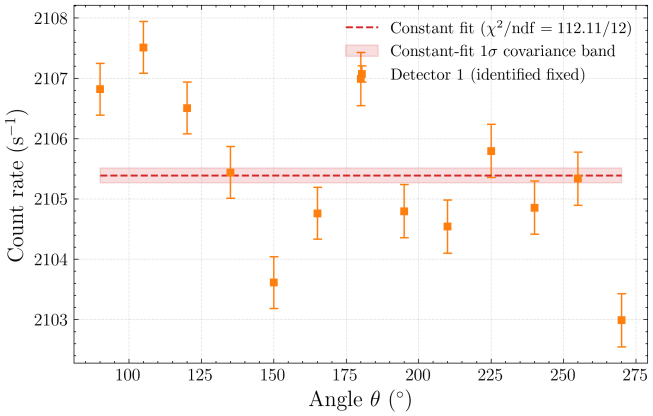


FIG. 31: Angle dependence of the fixed-detector singles rate with a weighted constant fit. Although the fixed detector remained at one physical angle throughout the measurement, the horizontal axis still labels the angle setting of the movable detector during the corresponding run. The shaded band indicates the 1σ covariance band of the constant fit.

The single rates shown in Figures 24–30 and discussed in Section VI B already show that the 60Co source is not perfectly centered. Figure 31 displays the rate of the fixed detector as a function of the scan angle. This does not mean that the fixed detector itself was moved. Rather, each point gives the count rate recorded by the

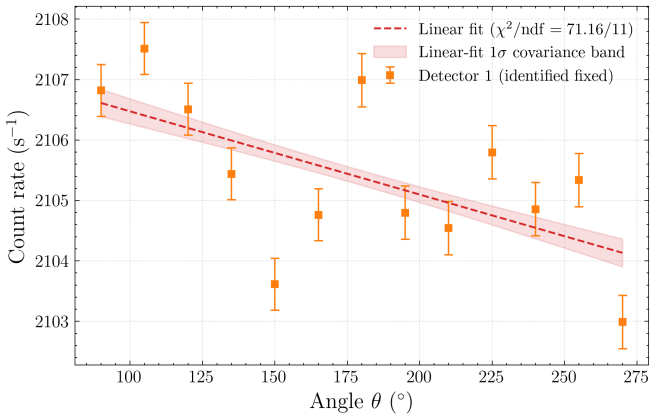


FIG. 32: Angle dependence of the fixed-detector singles rate with a weighted linear fit. The same data are shown as in Figure 31 , but now with a linear trend as an empirical description of the weak residual angle dependence. The shaded band indicates the propagated 1σ covariance band of the fit.

fixed detector in the run for which the movable detector was set to the corresponding angle. A weighted constant fit yields

$$K_0 = 2105.39(12)\,\mathrm{s}^{-1}, \qquad \chi^2/\mathrm{ndf} = 112.11/12, \quad (48)$$

so a strictly angle-independent fixed-detector rate is statistically rejected due its low p value. A weighted linear fit,

$$R_{\mathrm{fix}}(\theta) = a + b\theta, \quad (49)$$

gives

$$a = 2107.9(4)\,\mathrm{s}^{-1}, \quad (50)$$

$$b = -0.014(2)\,\mathrm{s}^{-1}\,{}^{\circ -1}, \quad (51)$$

with

$$\chi^2/\mathrm{ndf} = 71.16/11. \quad (52)$$

The linear model is therefore better than the constant one, but it is still not statistically compatible with the data. Quantitatively, however, the fixed-detector modulation remains relatively small: the rate varies only from $2102.99\,\mathrm{s}^{-1}$ to $2107.51\,\mathrm{s}^{-1}$, corresponding to a relative span of only 0.215 %.

By contrast, the movable-detector singles rate varies more strongly with angle and therefore carries the dominant signature of the misalignment, as shown in Figure 33 . Its rate changes from $1562.58\,\mathrm{s}^{-1}$ to $1621.91\,\mathrm{s}^{-1}$, corresponding to a relative span of 3.72 %. A source displaced from the center changes both the source–detector distance and the projected detector area as the movable detector is rotated. To first order in the dimensionless offset $\epsilon = \delta/R$, one therefore expects an approximately cosine-like modulation,

$$R_{\mathrm{mov}}(\theta) \approx R_0\,[1 + 2\epsilon\cos(\theta - \phi)] \quad (53)$$

The full geometric point-source model, derived in collaboration with Codex [6] and given in Appendix A 1 , refines this approximation to

$$R(\theta) = R_0 \frac{1 - \epsilon\cos(\theta - \phi)}{(1 + \epsilon^2 - 2\epsilon\cos(\theta - \phi))^{3/2}} \quad (54)$$

D. Geometric fit of the movable-detector singles rate

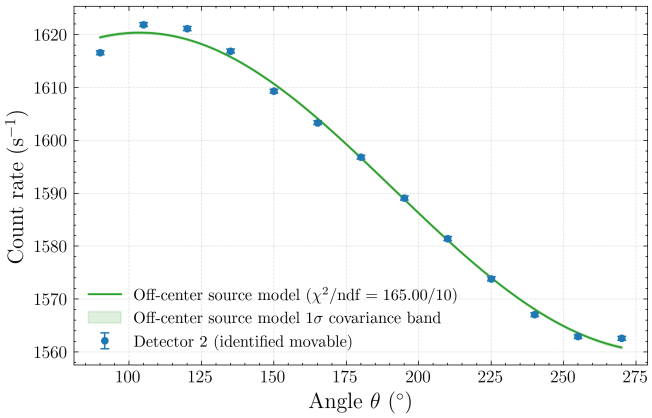


FIG. 33: Angle dependence of the movable-detector singles rate with the off-center source model. The solid curve shows the best fit of Equation (54) , and the shaded band indicates the propagated 1σ covariance band. The systematic modulation demonstrates that the setup is not perfectly centered.

Figure 33 shows the fit of Equation (54) to the movable-detector singles rate. The fitted parameters are

$$R_0 = 1589.8(2)\,\mathrm{s}^{-1}, \quad (55)$$

$$\epsilon = 0.009\,49(5), \quad (56)$$

$$\phi = 103.7(6)^{\circ}, \quad (57)$$

with

$$\chi^2/\mathrm{ndf} = 165.00/10 \quad (58)$$

Using the measured source–detector radius

$$R = 5.2(2)\,\mathrm{cm}, \quad (59)$$

this corresponds to an effective offset

$$\delta = \epsilon R = 0.0493(19)\,\mathrm{cm} = 0.493(19)\,\mathrm{mm}. \quad (60)$$

This number must not be overinterpreted. The counting uncertainties of the singles rates are fixed by Poisson statistics and are therefore not free fit parameters. The goodness-of-fit $\chi^2/\mathrm{ndf} = 165.00/10$ must therefore be attributed to model mismatch rather than to underestimated statistical noise. In other words, the model

captures the dominant scale of the misalignment visually, but not all details of the setup. Plausible missing effects are the finite detector size or imperfect detector pointing. A simpler cosine fit (shown in Equation (53)) was also tested for comparison, but it performs slightly worse in terms of its goodness-of-fit:

$$\chi^2/\mathrm{ndf} = 175.81/10, \tag{61}$$

so it does not improve the statistical compatibility either. The extracted offset should therefore be interpreted only as a phenomenological estimate of the misalignment scale, not as a precise determination of the true source displacement.

The same geometric effect also propagates into the coincidence rate. If the source is off-center, the acceptance of the movable detector changes with angle, and the measured coincidence rate inherits this modulation multiplicatively. In the idealized case of a perfectly constant fixed detector, one could remove this effect by dividing the coincidence rate by the movable-detector singles rate, up to an overall normalization. Here, however, the fixed detector is not perfectly constant either, even though its modulation is much weaker. We therefore corrected the coincidence rate using

$$R_{\mathrm{corr}}(\theta) = R_{3,\mathrm{sub}}(\theta)\,\frac{R_{\mathrm{fix}}(\theta)}{R_{\mathrm{mov}}(\theta)}, \tag{62}$$

where $R_{3,\mathrm{sub}}$ denotes the coincidence rate after subtraction of the random coincidence background per angle, $R_{\mathrm{fix}} = R_1$ is the fixed-detector single rate, and $R_{\mathrm{mov}} = R_2$ is the movable-detector single rate. Importantly, the random-coincidence subtraction was carried out angle by angle, because the dedicated random-coincidence measurement was performed at the same set of angles as the main angular-correlation scan.

Strictly speaking, the random-coincidence rate is angle dependent, since

$$R_{\mathrm{rand}}(\theta) = \Delta t_{\mathrm{res}} R_1(\theta) R_2(\theta) \tag{63}$$

inherits the angle dependence of the singles rates. Therefore, one common random-coincidence rate is not the formally correct description. In the present data, however, this dependence is very weak: the predicted random-coincidence rate ranges only from $0.106\,84\,\mathrm{s}^{-1}$ to $0.110\,44\,\mathrm{s}^{-1}$, corresponding to a maximal deviation of only about $0.0018\,\mathrm{s}^{-1}$ from the mean value $0.108\,79\,\mathrm{s}^{-1}$. This is negligible compared with the observed angular modulation of the coincidence rate. Hence a constant subtraction would have been an adequate approximation for this dataset. Nevertheless, since the random-coincidence measurement was performed at the same angles as the main scan, we used the more direct procedure and subtracted the measured random-coincidence rate individually at each angle before applying the misalignment correction.

E. Measurement of the angular correlation

For the final analysis, we combined all runs taken at the same angle by summing the total coincidence counts and total measuring times before forming the corresponding rates. The coincidence data were then corrected in two steps. First, we subtracted the measured random-coincidence rate R_{rand} angle by angle, using the dedicated random-coincidence measurement performed at the same set of angular positions as the main scan. Second, we compensated the geometric misalignment of the setup by scaling the coincidence rate with the ratio of the fixed-detector single rate R_{fix} and movable-detector single rate R_{mov}:

$$R_{\mathrm{corr}}(\theta) = [R_3(\theta) - R_{\mathrm{rand}}(\theta)]\,\frac{R_{\mathrm{fix}}(\theta)}{R_{\mathrm{mov}}(\theta)} \tag{64}$$

The uncertainties of the corrected rates were propagated from the counting statistics of all contributing quantities. In the subsequent fits, we used the full covariance matrix of the corrected angular-correlation data.

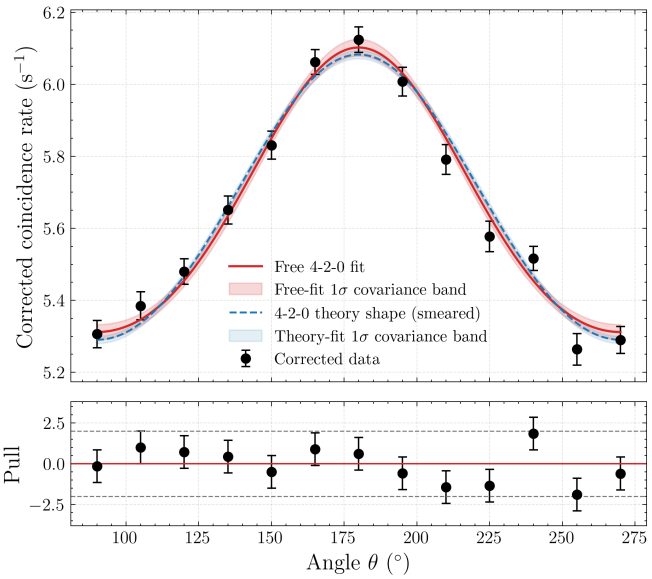


FIG. 34: Corrected angular-correlation data together with the free 4–2–0 Legendre fit and the normalized theoretical 4–2–0 shape. The red curve shows the free fit of the Legendre coefficients, while the blue dashed curve shows the 4–2–0 theory shape with literature coefficients fixed and only the normalization fitted. The shaded regions indicate the propagated 1σ covariance bands, and the lower panel shows the pulls of the free fit.

We first fitted the corrected angular-correlation data with the free Legendre expansion

$$W(\theta) = K_0\,[1 + A_{22}^{\mathrm{exp}} P_2(\cos\theta) + A_{44}^{\mathrm{exp}} P_4(\cos\theta)] \tag{65}$$

Here, K_0 is the normalization, while A_{22}^{exp} and A_{44}^{exp} denote the experimentally determined coefficients of the

second- and fourth-order Legendre terms. The fit yields

$$K_0 = 5.540(12)\,\mathrm{s}^{-1}, \tag{66}$$
$$A_{22}^{\mathrm{exp}} = 0.090(4), \tag{67}$$
$$A_{44}^{\mathrm{exp}} = 0.012(5), \tag{68}$$

with

$$\chi^2/\mathrm{ndf} = 14.64/10, \qquad p = 1.46 \times 10^{-1}. \tag{69}$$

We therefore conclude that the fit is likely statistically compatible with the corrected data, though we note residual effects might be present, which is why the p-value is off. Figure 34 shows that the fitted shape reproduces the measured angular dependence, with data points at 210°, 225°, 240°, and 255° not agreeing with the pull baseline within their 1σ interval. This could also be an explanation for the deviation from unity of χ^2ndf and the resulting p-value.

To account for the finite solid angle of the detectors, we corrected the measured Legendre coefficients according to

$$A_{kk}^{\mathrm{exp}} = Q_k^2 A_{kk}^{\mathrm{true}}. \tag{70}$$

The square of the attenuation factor Q_k appears because both detectors contribute a finite-angular-acceptance suppression of the anisotropy. We treated the two detectors as having the same attenuation factors Q_k, since no dedicated detector-specific determination of these factors was carried out. This is a simplifying assumption, not an independently verified property of the setup. Using the tabulated values $Q_2 = 0.9570$ and $Q_4 = 0.7114$ [2] , we obtain

$$A_{22}^{\mathrm{true}} = \frac{A_{22}^{\mathrm{exp}}}{Q_2^2} = 0.098(4), \tag{71}$$
$$A_{44}^{\mathrm{true}} = \frac{A_{44}^{\mathrm{exp}}}{Q_4^2} = 0.02(1). \tag{72}$$

The uncertainties were obtained by propagating the covariance matrix of the fitted coefficients through the solid-angle correction. Comparing these values with the literature expectations of $A_{22} \approx 0.1020$ and $A_{44} \approx 0.0091$ [3] shows that A_{22}^{true} agrees with the expected value within its 1σ interval, whereas A_{44}^{true} is off by a factor of 2.6.

The corrected Legendre coefficients may also be rewritten in the equivalent cosine form

$$W(\theta) = K_0' \left[1 + B\cos^2\theta + C\cos^4\theta\right]. \tag{73}$$

For the present data, the corresponding coefficients are

$$B = 0.06(4) \tag{74}$$
$$C = 0.11(4) \tag{75}$$

These uncertainties again include the covariance of the corrected Legendre coefficients.

As an additional test, we fitted the data with the theoretical 4–2–0 cascade shape, keeping the angular coefficients fixed to the literature values and varying only the normalization,

$$W_{4\text{–}2\text{–}0}(\theta) = K_0 \left[1 + Q_2^2 A_{22}^{4\text{–}2\text{–}0} P_2(\cos\theta) + Q_4^2 A_{44}^{4\text{–}2\text{–}0} P_4(\cos\theta)\right]. \tag{76}$$

With $A_{22}^{4\text{–}2\text{–}0} = 0.1020$ and $A_{44}^{4\text{–}2\text{–}0} = 0.0091$ [3] , the fit gives

$$K_0 = 5.54(1)\,\mathrm{s}^{-1}, \tag{77}$$

with

$$\chi^2/\mathrm{ndf} = 17.07/12, \qquad p = 1.47 \times 10^{-1}. \tag{78}$$

We see here that even with the literature values for the correlation coefficients A_{kk}, the p-value remains almost constant. We concluded that this can point to two things:

1. The corrected data is statistically compatible with the fixed theoretical 4–2–0 cascade shape, and we obtained a χ^2 value this extreme by pure chance or
2. the model mostly describes the data, however there is residual structure not captured by the model. This could come from contaminants in the source or potential electronic effects. However, we note that these are merely vague non-quantified hypotheses to explain the statistics obtained from the fit. We can therefore not with certainty conclude wether residual effects exists and if yes, what they are caused by.

Generally, visual inspection of Figure 34 shows that the free fit and the fixed theory shape lie almost on top of each other across the full angular range, showing that the free fit and the theory fit are consistent with each other within their covariance band. This is also evident by the statistical agreement of the two normalization constants K_0 from the free and theory constrained fit.

We also tested the alternative 2–1–0 cascade hypothesis using

$$W_{2\text{–}1\text{–}0}(\theta) = K_0 \left[1 + Q_2^2 A_{22}^{2\text{–}1\text{–}0} P_2(\cos\theta) + Q_4^2 A_{44}^{2\text{–}1\text{–}0} P_4(\cos\theta)\right] \tag{79}$$

with fixed coefficients [3]

$$A_{22}^{2\text{–}1\text{–}0} = 0.05, \qquad A_{44}^{2\text{–}1\text{–}0} = 0 \tag{80}$$

Only the normalization K_0 was fitted, yielding

$$K_0 = 5.60(1)\,\mathrm{s}^{-1}, \tag{81}$$

with

$$\chi^2/\mathrm{ndf} = 217.02/12, \qquad p = 9.87 \times 10^{-40} \tag{82}$$

This model is therefore decisively incompatible with the corrected angular-correlation data. Figure 35 shows that the 2–1–0 shape misses the measured peak height and side values by several standard deviations. The nearly invisible covariance band is simply a consequence of the tiny uncertainty on the fitted normalization and a "too

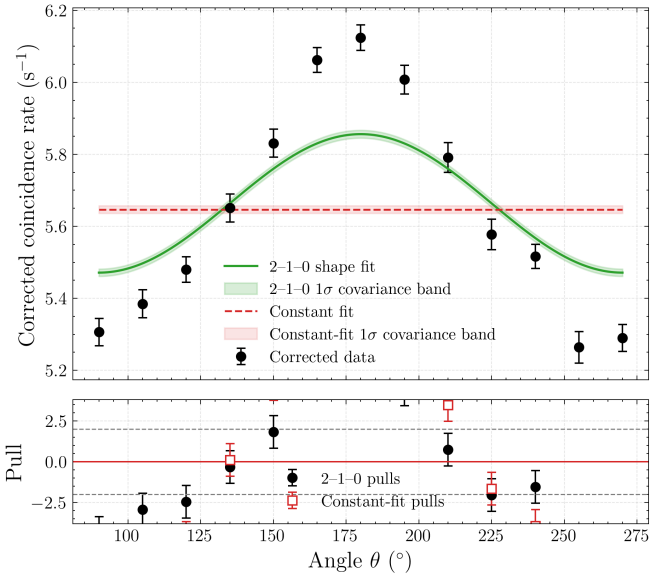


FIG. 35: Corrected angular-correlation data compared with the fixed 2–1–0 cascade shape and with a constant model. The green curve shows the 2–1–0 shape with only a free normalization, while the red dashed line shows the best constant fit. The covariance bands are extremely narrow because only one normalization parameter is fitted in both models; this does not indicate a good fit. Several pulls lie outside the displayed range, reflecting the strong incompatibility of both models with the data.

confident fitting algorithm"; it does not imply that the model describes the data well.

Finally, we tested a constant angular distribution as an isotropy hypothesis,

$$W(\theta) = K_0 \tag{83}$$

The fit gives:

$$K_0 = 5.65(1)\,\mathrm{s}^{-1} \tag{84}$$

with

$$\chi^2/\mathrm{ndf} = 774.50/12, \qquad p = 4.86 \times 10^{-158} \tag{85}$$

An isotropic coincidence rate is therefore excluded very strongly. Comparing the constant model with the free Legendre fit yields

$$\Delta\chi^2 = 759.86, \qquad \Delta\mathrm{ndf} = 2, \qquad p = 9.98 \times 10^{-166}, \tag{86}$$

which confirms that the observed angular dependence is highly statistically significant.

Overall, the final corrected data is clearly incompatible with both the 2–1–0 alternative and an isotropic distribution.

VII. CONCLUSION

We calibrated the full fast–slow coincidence chain of the setup and measured the angular γ-γ correlation of the ^{60}Co 4–2–0 cascade over the angular range from 90° to 270° in steps of 15°. The prompt-curve analysis yielded a resolving time of 32.3(10) ns, which was sufficiently short to limit random coincidences while still wide enough to retain the prompt coincidence plateau. A dedicated random-coincidence measurement was carried out at the same angular settings as the main scan, which allowed us to subtract the accidental background directly for each angle instead of relying on a global approximation.

The singles rates revealed that the setup was not perfectly centered as discussed in Sections VI B and VI C . In particular, the movable detector showed a clear angular modulation, while the fixed detector exhibited a smaller but still measurable variation. We therefore corrected the coincidence data with the ratio $R_{\mathrm{fix}}/R_{\mathrm{mov}}$. A simple off-center source model gave an effective displacement of $\delta = 0.49(2)$ mm, but the poor goodness-of-fit shows that this value should be interpreted only as a phenomenological estimate of the misalignment scale rather than as a precise geometric determination.

After random-coincidence subtraction and misalignment correction, the angular-correlation data are most statistically compatible with the expected 4–2–0 cascade shape. The free Legendre fit yields $A_{22}^{\mathrm{exp}} = 0.090(4)$ and $A_{44}^{\mathrm{exp}} = 0.012(5)$, which become $A_{22}^{\mathrm{true}} = 0.098(4)$ and $A_{44}^{\mathrm{true}} = 0.02(1)$ after finite-solid-angle correction. The coefficient A_{22}^{true} agrees well with the literature expectation, whereas A_{44}^{true} is less well constrained and remains larger by a factor of roughly 2.6 than the nominal value. At the same time, a fit with the fixed theoretical 4–2–0 shape remains statistically compatible with the free fit.

By contrast, both a constant angular distribution and the alternative 2–1–0 cascade shape are decisively excluded. The experiment therefore reproduces the essential anisotropic angular correlation of the ^{60}Co cascade. The dominant limitations are no longer the statistical counting precision alone, but rather residual geometric asymmetries, finite-solid-angle corrections, and other systematic effects that are not fully captured by the simplified correction models.

VIII. ACKNOWLEDGMENTS

We acknowledge the usage of Gemini 3.1 [7] for drafting support, text refinement, and LaTeX formatting assistance during the preparation of this manuscript as well as coding from Claude Code from Anthropic [8] and both coding and drafting support from Codex [6] from OpenAI.

Additionally, we note that all shown oscilloscope pictures of the output signals of the calibrated circuits are taken from [5] . We thank Nicolas Rück and Simon

Schöps for allowing us to do so.

Appendix A: Misalignment model

1. Derivation of the off-center source model

The following idealized model, derived in collaboration with Codex, was used to parameterize the angular dependence of the movable-detector singles rate used in Section VI C . The purpose of the model is not to reproduce every detail of the setup, but to describe the leading geometric effect of a source that is displaced from the nominal center of the detector circle.

We assume:

1. the ^{60}Co source can be treated as point-like,
2. the center of the movable detector moves on a circle of radius R,
3. the detector axis always points toward the nominal center of that circle,
4. the measured single rate is proportional to the solid angle subtended by the detector.

Under these assumptions, the only source of angular modulation is the offset of the source position away from the geometric center.

We choose a two-dimensional coordinate system in the detector plane. The center of the movable detector at scan angle θ is

$$\mathbf{r}_D(\theta) = R \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \tag{A1}$$

while the actual source position is displaced by a distance δ from the nominal center,

$$\mathbf{r}_S = \delta \begin{pmatrix} \cos\phi \\ \sin\phi \end{pmatrix}, \tag{A2}$$

where ϕ denotes the direction of the offset.

a. Source–detector distance. The vector from the detector center to the source is

$$\mathbf{r}_S - \mathbf{r}_D = \begin{pmatrix} \delta\cos\phi - R\cos\theta \\ \delta\sin\phi - R\sin\theta \end{pmatrix}. \tag{A3}$$

Its magnitude gives the source–detector distance,

$$r(\theta) = |\mathbf{r}_D - \mathbf{r}_S| = \sqrt{R^2 + \delta^2 - 2R\delta\cos(\theta - \phi)}. \tag{A4}$$

If $\delta = 0$, this reduces to the constant distance $r(\theta) = R$, as expected for a perfectly centered source.

b. Projected detector area. The detector does not only move closer to or farther away from the source. Since its axis always points toward the nominal center, an off-center source also changes the incidence angle of the radiation on the detector face. The unit normal of the detector, pointing toward the nominal center, is

$$\mathbf{n}(\theta) = -\begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}. \tag{A5}$$

The cosine of the incidence angle α is therefore

$$\cos\alpha = \frac{\mathbf{n}\cdot(\mathbf{r}_S - \mathbf{r}_D)}{r(\theta)}. \tag{A6}$$

Using the definitions above,

$$\mathbf{n}\cdot(\mathbf{r}_S - \mathbf{r}_D) = -\cos\theta\,(\delta\cos\phi - R\cos\theta) - \sin\theta\,(\delta\sin\phi - R\sin\theta) \tag{A7}$$

$$= R - \delta\cos(\theta - \phi), \tag{A8}$$

so that

$$\cos\alpha = \frac{R - \delta\cos(\theta - \phi)}{r(\theta)}. \tag{A9}$$

c. Count-rate model from the subtended solid angle. For an isotropic point source, the detected singles rate is proportional to the solid angle subtended by the detector face. For a small detector area A, this solid angle scales as

$$\Omega(\theta) \propto \frac{A\cos\alpha}{r(\theta)^2}. \tag{A10}$$

Since A is constant, the count rate may be written as

$$R(\theta) \propto \frac{\cos\alpha}{r(\theta)^2}. \tag{A11}$$

Substituting Equations (A4) and (A9) yields

$$R(\theta) = C\,\frac{R - \delta\cos(\theta - \phi)}{\left(R^2 + \delta^2 - 2R\delta\cos(\theta - \phi)\right)^{3/2}}, \tag{A12}$$

where C is an overall normalization constant.

d. Dimensionless form. It is convenient to introduce the dimensionless offset

$$\epsilon = \frac{\delta}{R}. \tag{A13}$$

Factoring out the appropriate powers of R from Equation (A12) gives

$$R(\theta) = R_0\,\frac{1 - \epsilon\cos(\theta - \phi)}{\left(1 + \epsilon^2 - 2\epsilon\cos(\theta - \phi)\right)^{3/2}}, \tag{A14}$$

with

$$R_0 = \frac{C}{R^2}. \tag{A15}$$

Equation (A14) is the form used in the fit in Figure 33 .

e. Small-offset limit. For $\epsilon \ll 1$, the model reduces to an approximately cosine-like modulation. Expanding the denominator to first order,

$$\left(1+\epsilon^2-2\epsilon\cos(\theta-\phi)\right)^{-3/2} = 1+3\epsilon\cos(\theta-\phi)+\mathcal{O}(\epsilon^2), \tag{A16}$$

so that

$$R(\theta) = R_0\left(1-\epsilon\cos(\theta-\phi)\right)\left(1+3\epsilon\cos(\theta-\phi)\right)+\mathcal{O}(\epsilon^2) \tag{A17}$$

$$= R_0\left[1+2\epsilon\cos(\theta-\phi)\right]+\mathcal{O}(\epsilon^2). \tag{A18}$$

Thus, the off-center model naturally predicts a cosine-like angular dependence to leading order, with maximum rate near the direction $\theta \simeq \phi$ in which the detector is closest to and most directly facing the displaced source.

f. Interpretation and limitations. The parameter ϵ measures the offset in units of the detector-circle radius, while ϕ specifies the direction of that offset. If the source were perfectly centered, one would have $\epsilon = 0$ and therefore

$$R(\theta) = R_0, \tag{A19}$$

that is, an angle-independent singles rate.

The model is deliberately simplified. It neglects finite detector size, finite source size, scattering from surrounding material, imperfect detector pointing, absorption in holders or collimators, and possible long-term drifts of the electronics.

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